

Hugo HADJUR, PhD

Junior AI & Statistical Methodologist

Grenoble, FRANCE — hugo.hadjur@saryga.com

PhD in computer science and artificial intelligence. Multicultural background from professional experiences in **France, Japan and the US**, involving **academic research**, **teaching** and working in the **industry**.



Thesis manuscript



Academic Page



Google Scholar



LinkedIn

WORK EXPERIENCE

SINCE DEC. 2023

CDI



SARYGA
STATISTICS & METHODOLOGY

Junior AI & Statistical Methodologist at **Saryga** (FRANCE)

— **Biostatistics and Pharma projects:** predictive biomarker cutoff selection methods; survival analysis for PDX; quantitative decision-making methods at the portfolio level; implementation and simulation of PoC multi-arm trials, optimization of sample size.

— **Tech projects:** semantic analysis using LLMs, zero-shot text classification, paraphrase mining; development of R Shiny web app to identify targets using Multi-Criteria Decision-Making methods.

SEPT. 2020 - AUG. 2023

3-year CDD



ENS DE LYON

aivancity

PARIS-CACHAN

PhD Candidate and Assistant Professor at **École normale supérieure de Lyon** & **aivancity** (FRANCE)

— PhD Research Topic: **Designing sustainable, autonomous, and energy efficient Internet of Things systems**, applied to smart beehives

PhD under the advisorship of **Laurent Lefèvre** and **Doreid Ammar** (LIP-Avalon laboratory)

— Teaching courses: **advanced data analysis and visualization**, **data preprocessing and visualization** and **coding for AI and data science** (Python, Tableau)

Head of online course platform **aivancityX**

SEP. 2019 - AUG. 2020

& OCT. 2018 - DEC. 2018

1-year CDD



Instructor & Researcher at **emlyon Business School** (FRANCE)

— Research: Connected beehives, bees' data analysis (**IoT**, **machine learning**, **computer vision**)

— Teaching: **Python programming bootcamp** and **business analyst toolbelt** (SQL, Excel, Tableau)

JAN. 2019 - SEP. 2019

8-month VIE



Data Analyst at **Schneider Electric**, Houston, TX (USA)

— Supported the business by building **reports/dashboards and KPIs** to monitor performance

— Conducted **database integration** to form practical recommendations

— **Communicated findings** and insights to cross-functional team members and management

— Tool used: **SQL**, **Tableau**, **Python**

OCT. 2017 - SEP. 2018

1-year exchange



Student Researcher at **Kyoto University's Ishii Lab** (JAPAN)

— **Master's thesis:** master board games thanks to **Reinforcement Learning** (Q-Learning, SARSA, Deep Reinforcement Learning)

MAY-SEP. 2017

4-month internship



Data Science Research Internship at **CiNet** (Center for Information and Neural Networks, Osaka, JAPAN)

— **Conducted research** about the links between **Twitter data**, human behavior and brain data

— **Analyzed large data sets** ($\approx 500,000$ Twitter users) thanks to **PCA**, **clustering & regression** techniques

— Worked in a **Japanese environment**

PUBLICATIONS

Hugo Hadjur, Doreid Ammar, Laurent Lefèvre. **Deep Reinforcement Learning for Energy-efficient Selection of Embedded Services at the Edge**, 2024 IEEE International Conferences on Internet of Things (iThings), Copenhagen, Denmark, 2024, pp. 67-74, [10.1109/iThings-GreenCom-CPSCoM-SmartData-Cybermatics62450.2024.00034](#). [hal-04708697](#)

Hugo Hadjur, Doreid Ammar, Laurent Lefèvre. **Services Orchestration at the Edge and in the Cloud for Energy-Aware Precision Beekeeping Systems**, 2023 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW), St. Petersburg, FL, USA, 2023, pp. 769-776, [10.1109/IPDPSW59300.2023.00129](#). [hal-04091575](#)

Hugo Hadjur, Doreid Ammar, Laurent Lefèvre. **Toward an intelligent and efficient beehive: A survey of precision beekeeping systems and services**, Computers and Electronics in Agriculture, Elsevier, 2022, 192, pp. 1-16. [10.1016/j.compag.2021.106604](#). [hal-03483914v2](#)

Kazuma Mori, **Hugo Hadjur**, Masahiko Haruno. **Natural Language Content Mediates the Association Between Active Interactions on Social Network Services and Subjective Well-Being**, Cyberpsychology, Behavior, and Social Networking, 2022, pp. 678-685. [10.1089/cyber.2021.0340](#)

Hugo Hadjur, Doreid Ammar, Laurent Lefèvre. **Analysis of Energy Consumption in a Precision Beekeeping System**. IoT '20 - 10th International Conference on the Internet of Things, Oct 2020, Malmö, Sweden. pp. 1-11, [10.1145/3410992.3411010](#). [hal-02973772](#)

COMMUNICATION

POSTERS

— **ISCB** - Thessaloniki, Greece, 2024. [Simple Approaches for Portfolio Quantitative Decision-Making](#)

PRESENTATIONS

— **GreenDays 2023** - Lyon, France, 2023. Optimiser la sélection de tâches sous contrainte énergétique dans des systèmes IoT grâce à l'apprentissage par renforcement

— **Journées du GDR Réseaux et Systèmes Distribués** - Rennes, France, 2022. Conception de systèmes connectés durables, autonomes et à basse consommation énergétique pour un réseau de ruches connectées distribuées

— **IoT'20** - Online, 2020. Analysis of Energy Consumption in a Precision Beekeeping System

PROGRAMMING & SOFTWARE

PROGRAMMING:	Expert knowledge: Python (data analysis, visualization and deep learning: time series, computer vision and reinforcement learning), SQL , Tableau , R Advanced: Javascript (full-stack), Java , C , C++ , MATLAB Experience using: Rest APIs , Git
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DESIGN & TEXT:	L^AT_EX , Adobe Photoshop , Microsoft Office , Inkscape
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2020-2023	PhD at ENS de Lyon (LIP-Avalon laboratory) & aivancity
	Develop and model Internet of Things (IoT) systems and services based on Artificial Intelligence (AI) to support beekeepers in their work and preserve bees while studying the efficiency of connected beehives that are autonomous, sustainable, distributed and operate under limited energy budget.
2017-2018	Exchange year at Kyoto University's Graduate School of Informatics
	Alongside courses in English and Japanese, I worked toward my Master's thesis. This thesis proposes a reinforcement learning method applied to the game of bridge. The Q-Learning and SARSA techniques reach a level of play comparable to an intermediate human player. — Classes in English & Japanese in the department of social informatics: statistics, machine learning
2015-2017	French “Grandes Ecoles”: Ensimag, Grenoble Institute of Technology (equivalent: Master's Degree)
	— MMIS specialization: programming, advanced statistics, data mining, database manipulation
2013-2015	Scientific preparatory classes at CPP Grenoble (equivalent: Bachelor's Degree)
	— 2-year intensive university-level preparatory course to enter highly competitive Ivy League engineering schools
2007-2013	Middle school and high school in Tokyo (4 years at LFIT) and in Singapore (1 year at LFS)
 	— Scientific baccalaureate with highest honours (French school)